INFOSYS SE PSEUDOCODE VAULT • RECURSION & FUNCTION CALL STACKS

Recursive Call Trees, Base Cases, Fibonacci, GCD, Stack Trace • Total Solved Questions: 12

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Function Recursive-style

Q1. What will be the output of the following pseudocode?

```
Integer f(Integer n)
if (n <= 1)
return 1
end-if
return n + f(n-2)
End function
display f(7)
```

(A) 10
(B) 12
(C) 16
(D) 18

Correct Answer: Option (C) : 16

■ **Step-by-Step Logic:** Execution Trace: 1. Identify base cases for termination. 2. Trace recursive function call stack and branch expansions. 3. Evaluate leaf returns and unwind the call stack bottom-up. 4. Summing/accumulating the return values yields Option (C) (16).

[OnlineStudy4U Master E-Book 4]

Topic: Pseudocode Execution Tracing

Q2. What will be the output of the following pseudocode?

```
What value is printed by the following
Pseudo Code?
Function
Mystery(n)
if
n <= 1 then
return
1
end
if
return
n * Mystery(n - 1)
End
Function
Print
Mystery(5)
```

(A) 15
(B) 24
(C) 120
(D) 125

Correct Answer: Option (C) : 120

■ **Step-by-Step Logic:** Mystery(n) computes n!. Mystery(5) = 5 × 4 × 3 × 2 × 1 = 120.

[OnlineStudy4U Master E-Book 4]

Topic: Pseudocode Execution Tracing

Q3. What will be the output of the following pseudocode?

```
What value is printed by the following
Pseudo Code?
Function
F(n)
if
n == 0 then return 0
if
n == 1 then return 1
return
F(n - 1) + F(n - 2)
End
Function
Print
F(6)
```

(A) 5
(B) 8
(C) 13
(D) 21

Correct Answer: Option (B) : 8

■ **Step-by-Step Logic:** F is the Fibonacci sequence: F(0)=0, F(1)=1, F(2)=1, F(3)=2, F(4)=3, F(5)=5, F(6)=8. So the printed value is 8.

[OnlineStudy4U Master E-Book 4]

Topic: Pseudocode Execution Tracing

Q4. What will be the output of the following pseudocode?

```
[Recursion]
What does S(5) return?
Function
S(n)
if
n == 0 then return 0
return
n + S(n - 1)
End
Function
Print
S(5)
```

(A) 10
(B) 15
(C) 16
(D) 25

Correct Answer: Option (B) : 15

■ **Step-by-Step Logic:** S(n) is the sum 5+4+3+2+1+0 = 15.

[OnlineStudy4U Master E-Book 4]

Topic: Pseudocode Execution Tracing

Q5. What will be the output of the following pseudocode?

```
[Recursion]
What does P(4) return?
Function
P(n)
if
n == 1 then return 2
return
P(n - 1) * 2
End
Function
Print
P(4)
```

(A) 8
(B) 12
(C) 16
(D) 24

Correct Answer: Option (C) : 16

■ **Step-by-Step Logic:** P(1)=2, P(2)=4, P(3)=8, P(4)=16.

[Campus Monk Day 04 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q6. What will be the output of the following pseudocode?

```
1.     Find the output of the following pseudo code.
Integer n
Set n = 11
Integer arr[4] = { 10, 11, 1, 2 }
arr[1]=arr[1]-arr[3]
arr[2]=arr[2]+arr[2]
arr[3]=1+arr[3]
n=arr[3]
if(n>arr[2])
n=n-1
Else
n=n-1
End if

Print n
```

(A) 7
(B) 18
(C) 22
(D) 2

Correct Answer: Option (D) : 2

■ **Step-by-Step Logic:** Execution Trace: 1. Identify base cases for termination. 2. Trace recursive function call stack and branch expansions. 3. Evaluate leaf returns and unwind the call stack bottom-up. 4. Summing/accumulating the return values yields Option (D) (2).

[Campus Monk Day 08 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q7. What will be the output of the following pseudocode?

```
Q7.what is the output for number=4 int something(int
number) { if(number <= 0) return 1; else return number *
something(number-1); } something(4);
```

(A) 12
(B) 24
(C) 1
(D) 20

Correct Answer: Option (B) : 24

■ **Step-by-Step Logic:** Execution Trace: 1. Identify base cases for termination. 2. Trace recursive function call stack and branch expansions. 3. Evaluate leaf returns and unwind the call stack bottom-up. 4. Summing/accumulating the return values yields Option (B) (24).

[College ITM Mock Paper 1]

Topic: Infosys SE College Mock Test

Q8. What will be the output of the following pseudocode?

```
What is printed?
int f(int n) {
        if (n <= 1) return n;
        return f(n-1) + 2 * f(n-2);
}
print(f(6))
```

(A) 15
(B) 21
(C) 27
(D) 31

Correct Answer: See Logic

■ **Step-by-Step Logic:** Execution Trace: 1. Identify base cases for termination. 2. Trace recursive function call stack and branch expansions. 3. Evaluate leaf returns and unwind the call stack bottom-up. 4. Summing/accumulating the return values yields .

[College ITM Mock Paper 2]

Topic: Infosys SE College Mock Test

Q9. What will be the output of the following pseudocode?

```
What is printed?
int f(int n) {
        if (n == 0) return 1;
        if (n == 1) return 2;
        return f(n-1) + f(n-2) * 2;
}
print(f(5))
```

(A) 26
(B) 30
(C) 32
(D) 34

Correct Answer: See Logic

■ **Step-by-Step Logic:** Execution Trace: 1. Identify base cases for termination. 2. Trace recursive function call stack and branch expansions. 3. Evaluate leaf returns and unwind the call stack bottom-up. 4. Summing/accumulating the return values yields .

[College ITM Mock Paper 4]

Topic: Infosys SE College Mock Test

Q10. What will be the output of the following pseudocode?

```
What is printed? (% is the modulo operator)
int gcd(int a, int b) {
        if (b == 0) return a;
        return gcd(b, a % b);
}
print(gcd(252, 105))
```

(A) 21
(B) 42
(C) 7
(D) 14

Correct Answer: See Logic

■ **Step-by-Step Logic:** Execution Trace: 1. Identify base cases for termination. 2. Trace recursive function call stack and branch expansions. 3. Evaluate leaf returns and unwind the call stack bottom-up. 4. Summing/accumulating the return values yields .

[College ITM Mock Paper 5]

Topic: Infosys SE College Mock Test

Q11. What will be the output of the following pseudocode?

```
What is printed? (This computes the number of grid paths
from one corner to another, moving only
right or down.)
int f(int n, int m) {
        if (n == 0 || m == 0) return 1;
        return f(n-1, m) + f(n, m-1);
}
print(f(3, 2))
```

(A) 10
(B) 8
(C) 12
(D) 9

Correct Answer: See Logic

■ **Step-by-Step Logic:** Execution Trace: 1. Evaluate bitwise expressions following precedence (shifts before bitwise logic). 2. Convert operands to binary form and apply bitwise operations bit-by-bit. 3. Perform remaining arithmetic updates on the accumulated values. 4. The final printed output evaluates to .

[College ITM Mock Paper 6]

Topic: *Infosys SE College Mock Test*

Q12. What will be the output of the following pseudocode?

```
What is printed? (/ is integer division, % is modulo)
int f(int n) {
        if (n <= 1) return n;
        if (n % 2 == 0) return f(n / 2);
        return f(n - 1) + f(n + 1);
}
print(f(11))
```

(A) 5
(B) 4
(C) 6
(D) 3

Correct Answer: See Logic

■ **Step-by-Step Logic:** Execution Trace: 1. Identify base cases for termination. 2. Trace recursive function call stack and branch expansions. 3. Evaluate leaf returns and unwind the call stack bottom-up. 4. Summing/accumulating the return values yields .